NATIONAL UNIVERSITY OF LESOTHO

EE3404 Circuits I

Critical-Thinking Quizzes, Pair Activities and Individual Assignments

Capacitors • Inductors • Dependent Sources • Kirchhoff's Voltage Law

Prepared for classroom use • 2026

Part A: Ten pair quizzes

Quiz 1. Can a capacitor voltage jump?

Concept prediction • Capacitor continuity

A 100 μF capacitor has $v_C(0-) = 6$ V. A classmate claims that at $t = 0$ an ideal switch can make $v_C(0+) = 0$ V instantaneously.

- Decide whether the claim is valid and justify your answer from $i_C = C(dv_C/dt)$.
- If the change occurred uniformly in 1 μs, calculate the required average current and discuss what the result implies about an ideal instantaneous jump.
- State one non-ideal mechanism through which a real capacitor voltage may appear to change very rapidly.

Suggested time: 8 minutes

Quiz 2. Energy is not the same as capacitance

Quantitative reasoning • Stored energy

Two capacitors are available: A = 220 μF rated at 16 V and B = 100 μF rated at 25 V.

- Calculate the maximum ideal stored energy of each using $w_C = \frac{1}{2}Cv^2$.
- Which stores more energy at its rated voltage? Explain why comparing capacitance alone gives the wrong conclusion.
- If both are operated at 12 V, repeat the comparison and identify the design trade-off omitted by this ideal calculation.

Suggested time: 10 minutes

Quiz 3. Designing a parallel-plate capacitor

Geometry • Permittivity • Sensitivity

A parallel-plate capacitor uses dielectric $\varepsilon_r = 4.0$, plate area $A = 60\ \text{cm}^2$ and separation $d = 0.50$ mm. Take $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m.

- Calculate $C = \varepsilon_r\varepsilon_0 A/d$, carefully converting all quantities to SI units.
- Without recalculating from scratch, predict and quantify the percentage change in C if A increases by 10% while d decreases by 10%.
- Debate which manufacturing tolerance, area or spacing, is likely to be more difficult to control and why.

Suggested time: 10 minutes

Quiz 4. Reading a non-constant capacitor current

Calculus • Piecewise signals

A 50 μF capacitor carries $i_C(t) = 0.20t$ A for $0 \le t \le 4$ s and has $v_C(0) = -2$ V.

- Derive $v_C(t)$ from $v_C(t) = v_C(0) + (1/C)\int_0^t i_C(\tau)d\tau$.
- Calculate $v_C(4\text{ s})$ and the change in stored energy.
- One partner checks the result by differentiating $v_C(t)$; the other checks dimensions and physical direction of energy flow.

Suggested time: 12 minutes

Quiz 5. Inductor voltage and current continuity

Inductor dynamics • Interpretation

A 40 mH inductor has $i_L(t) = 3e^{(-50t)}$ A for $t \ge 0$, referenced into the positive-voltage terminal.

- Find $v_L(t) = L(di_L/dt)$ and $p_L(t) = v_L(t)i_L(t)$.
- Explain the negative sign of $p_L(t)$: is the inductor absorbing or delivering energy?
- Verify that the total energy delivered from $t = 0$ to ∞ equals $\frac{1}{2}Li_L^2(0)$.

Suggested time: 12 minutes

Quiz 6. Four dependent sources, four different units

Controlled sources • Dimensional analysis

For the source relations $v_o = \mu v_x$, $v_o = r i_x$, $i_o = g v_x$ and $i_o = \beta i_x$, classify each as VCVS, CCVS, VCCS or CCCS.

- State the SI units of μ, r, g and β.
- For $v_x = 20$ mV, $i_x = 2$ mA, $\mu = 80$, $r = 1.5$ kΩ, $g = 0.25$ S and $\beta = 40$, calculate all four source outputs.
- Explain why a dependent source is not “passive” merely because it depends on another circuit variable.

Suggested time: 10 minutes

Quiz 7. KVL: disagreement without contradiction

KVL • Sign convention • Critical comparison

A loop contains a 24 V source (rise in the chosen clockwise traversal), a 6 Ω resistor and a dependent voltage source of value 3i V opposing the 24 V source. The clockwise current is i.

- Write and solve the clockwise KVL equation.
- Reverse the traversal direction, write a second KVL equation, and show that it gives the same current.
- Explain precisely why different-looking KVL equations can both be correct.

Suggested time: 10 minutes

Quiz 8. Is every loop equation independent?

Topology • Linear dependence

A planar circuit has two meshes sharing one branch. A student writes KVL around the left mesh, right mesh and outer perimeter, then assumes all three equations are independent.

- Use algebraic reasoning to show why the outer-loop equation can be formed from the two mesh equations.
- State a practical consequence of using a redundant equation in a simultaneous-equation solution.
- Propose a reliable rule for choosing a sufficient set of independent loops.

Suggested time: 8 minutes

Quiz 9. Solar lantern ride-through

Applied scenario • Capacitor sizing • Engineering judgement

A solar lantern controller draws a constant 120 mA. During a battery changeover, a capacitor must keep the controller voltage from falling below 4.5 V from an initial 5.0 V for 80 ms. Ignore converter losses and equivalent series resistance (ESR).

- Use $i = C|\Delta v|/\Delta t$ to find the minimum capacitance.
- Calculate the energy released over the interval and compare it with load energy approximated using the average voltage.
- Recommend a practical capacitance above the ideal minimum and identify two real effects that justify the margin.

Suggested time: 12 minutes

Quiz 10. Pressure analogy: useful but incomplete

Analogy • Model limitations • Discussion

The lecture compares voltage with pressure, current with fluid flow, KCL with flow conservation and KVL with pressure changes around a closed hydraulic path.

- Give one way the analogy helps explain KVL and one way it can mislead students about capacitor or inductor behaviour.
- Map a hydraulic storage tank and a flywheel-like fluid inertia to capacitor and inductor concepts.
- Decide whether analogy or equations should dominate an engineering design decision, and defend your position.

Suggested time: 8 minutes

Part B: Five extended classroom questions for pairs

Classroom 1. Faulty phone-charger diagnosis

Applied scenario • KVL • Dependent source • Power balance

A simplified 5 V USB regulator is modelled by an ideal dependent voltage source $v_o = 12v_x$ in series with an internal resistance $R_s = 0.40\ \Omega$. A feedback divider gives $v_x = 0.08v_o$. The load draws $I_L = v_o/R_L$ with $R_L = 5.0\ \Omega$. A 9 V input source supplies the regulator stage.

- Discuss whether $v_o = 12v_x$ and $v_x = 0.08v_o$ alone uniquely determine a non-zero output. What does their product imply?
- Construct a consistent KVL-based equivalent model that includes the 9 V input and R_s, clearly stating any additional controlled-source relation you introduce.
- Solve for output voltage, load current and load power for your model.
- Explain how feedback gain, internal resistance and loading affect regulation; identify what the simplified model cannot predict.

Suggested time: 20 minutes • Emphasis: modelling assumptions, not one memorised topology

Classroom 2. Two ways to obtain the same equivalent capacitance

Series/parallel capacitors • Charge/voltage reasoning

Capacitors $C_1 = 6\ \mu F$ and $C_2 = 3\ \mu F$ are connected in series across 18 V.

- Derive the equivalent capacitance using charge equality and KVL rather than quoting the formula.
- Find the charge and voltage across each capacitor; verify KVL.
- Reconnect them in parallel across 18 V. Calculate the new total energy and compare it with the series case.
- Explain where energy must go if an ideal switch performs the reconnection after both capacitors were charged in series.

Suggested time: 20 minutes

Classroom 3. Sensor-controlled current source

Applied electronics • VCCS • Energy and ratings

A light sensor produces $v_x = 25$ mV per klux. A VCCS drives an LED string with $i_o = gv_x$, where $g = 0.80$ S. The source compliance voltage is 12 V and the LED string is approximated as a constant 9 V load.

- Find i_o and source output power at illuminances of 0.2, 0.5 and 1.0 klux.
- If the LED current must not exceed 20 mA, determine the maximum permissible illuminance or redesign g.
- Discuss why the VCCS model must include current limiting and compliance voltage to represent a realizable circuit.
- Sketch the expected i_o-versus-illuminance characteristic including saturation.

Suggested time: 20 minutes

Classroom 4. Capacitor versus inductor for short energy support

Comparative design • Scaling laws

A 10 W electronic load requires 0.50 J of short-duration support. Candidate storage elements are a capacitor operated from 12 V down to 9 V and an inductor whose current may fall from 5 A to 2 A.

- Find the required C from $\Delta w = \tfrac{1}{2}C(V1^2 - V2^2)$.
- Find the required L from $\Delta w = \tfrac{1}{2}L(I1^2 - I2^2)$.

- Calculate the ideal support time at 10 W.
- Compare voltage/current stress, control requirements, size, losses and safety. Select an element for a DC electronics application and justify the choice.

Suggested time: 20 minutes

Classroom 5. Find and repair the reasoning error

Error analysis • KVL • Passive sign convention

A student analyses a loop containing a 15 V source, a 5 Ω resistor and a 2 H inductor. With clockwise i and the source driving the loop, the student writes 15 + 5i + 2(di/dt) = 0 and concludes that a zero-current inductor must initially develop −15 V.

- Draw a consistent set of polarities and repair the KVL equation.
- For i(0) = 0, solve the differential equation and calculate i(0.20 s).
- Find vL(0+) and vL(∞), and explain both physically.
- Identify whether the original error was algebraic, directional, conceptual, or a combination.

Suggested time: 25 minutes

indicative answers and discussion cues

These are concise checkpoints, not exhaustive solutions. defensible alternatives with explicit assumptions will be rewarded.

Quiz 1: $v_C(0+) = 6$ V for finite current. Uniform 6 V decrease in 1 μs requires $i = C\Delta v/\Delta t = -600$ A. Parasitic inductance, ESR, breakdown or measurement bandwidth may shape a rapid observed change.

Quiz 2: At ratings: A = 0.02816 J; B = 0.03125 J, so B stores more. At 12 V: A = 0.01584 J; B = 0.00720 J. Omitted trade-offs include ESR, ripple current, lifetime, tolerance and derating.

Quiz 3: C ≈ 425 pF. New/old = 1.10/0.90 = 1.222..., hence +22.2%. Spacing often strongly affects process consistency and breakdown.

Quiz 4: $v_C(t) = -2 + 2000t^2$ V; $v_C(4) = 31{,}998$ V. The extreme result is intentional: it exposes the combination of a small C, long duration and rising current; $\Delta w \approx 25.6$ kJ.

Quiz 5: $v_L = -6e^{-50t}$ V; $p_L = -18e^{-100t}$ W. Energy delivered is $\int 18e^{-100t}dt = 0.18$ J = ½(0.04)(3^2).

Quiz 6: VCVS μ dimensionless → 1.6 V; CCVS r in Ω → 3 V; VCCS g in S → 5 mA; CCCS β dimensionless → 80 mA. A controlled source can exchange energy supplied by an unshown power supply.

Quiz 7: One convention gives $24 - 6i - 3i = 0$, hence $i = 8/3$ A. Reversing traversal multiplies the entire equation by −1.

Quiz 8: With consistent branch orientations, outer KVL is the sum/difference of mesh KVL equations; the shared-branch terms cancel. Redundancy can produce a singular system.

Quiz 9: $C_{min} = 0.12(0.08)/0.5 = 0.0192$ F = 19.2 mF. Released energy = ½C($5^2 - 4.5^2$) = 45.6 mJ; load estimate = 0.12×4.75×0.08 = 45.6 mJ. A practical choice might be 22–33 mF before derating; account for ESR, leakage, tolerance and temperature.

Quiz 10: Useful: potential changes sum around a closed path. Limits: hydraulic pictures can obscure electromagnetic fields, displacement current and sign/reference choices. Equations govern design; analogy supports intuition.

Classroom 2: Series $C_{eq} = C_1C_2/(C_1+C_2) = 2$ μF; Q = 36 μC; voltages 6 V and 12 V. Series energy = 0.324 mJ. Parallel energy at 18 V = ½(9 μF)(18^2) = 1.458 mJ.

Classroom 3: Sensor outputs: v_x = 5, 12.5, 25 mV; i_o = 4, 10, 20 mA; power at 9 V = 36, 90, 180 mW. At g = 0.8 S, 1 klux gives 20 mA.

Classroom 4: $C = 2(0.5)/(12^2 - 9^2) \approx 15.87$ mF. $L = 2(0.5)/(5^2 - 2^2) \approx 47.62$ mH. Ideal support time = 0.05 s.

Classroom 5: Consistent KVL: $15 - 5i - 2di/dt = 0$. $\tau = L/R = 0.4$ s; $i = 3(1-e^{-t/0.4})$ A; $i(0.2) \approx 1.180$ A; $v_L(0+) = 15$ V and $v_L(\infty) = 0$.

Source alignment

- EEE202 Lecture 2, “Capacitors; Inductors; Dependent Sources; KVL”: energy-storage elements; $q = Cv$; $i = Cdv/dt$; capacitor energy; $v = Ldi/dt$; source classifications; KVL and polarity; physical analogies.
- J. D. Irwin and R. M. Nelms, Basic Engineering Circuit Analysis, 11th ed.: Ch. 1 (dependent sources), Ch. 2 (Kirchhoff’s laws and KVL), Ch. 6 (capacitors, inductors and combinations).
- C. K. Alexander and M. N. O. Sadiku, Fundamentals of Electric Circuits, 7th ed.: Ch. 1 (dependent sources), Ch. 2 (basic laws/KVL), Ch. 6 (capacitors and inductors).